# PSTAT 274 Final Project

Time Series Analysis and Forecasting of Hotel Booking Demand

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## Abstract

The project aims to forecast future hotel booking volumes by conducting a time series analysis of hotel booking demand. The dataset used in this study has a total of 119,390 booking records that span over two years, from July 2015 to August 2017. Daily data were aggregated into weekly data instead of monthly data to ensure the data sample size is sufficient. Two methods, the Box-Jenkins approach (SARIMA(p, d, q)x(P, D, Q)) and a machine learning approach (Neural Network Autoregression), were used in this project. ARIMA(0,1,2)(0,1,0)[52] was identified as the optimal model; it successfully captured the non-stationary trends and the annual seasonality of the data and predicted a significant rise in bookings for the upcoming summer that matches the expectations. In contrast, the NNAR model generated a different result from the SARIMA model, likely due to the limitations of the relatively short sampling window. The findings demonstrate that the traditional parametric models have advantages over unconstrained machine learning models when handling high-frequency seasonal data, highlighting their value and utility in the hotel industry.

# 1. Introduction

People who love traveling would know that during holidays, the prices of everything related to the tourism industry, such as theme park tickets and airfare, would rise significantly, and hotels are no exception to this rule. In the hotel industry, demand is heavily driven by holidays and seasonal tourism. Given that most holidays fall on fixed dates each year, we can roughly infer that booking data in the hotel industry exhibits cyclical patterns. It is crucial to correctly predict the booking time of the guests because it's helpful for optimizing revenue management, staffing, and resource allocation. The purpose of this project is to use the historical hotel booking data to build models that can forecast the volume of weekly reservations for a 12-week horizon.

I choose this topic to explore how seasonal trends impact commercial revenue; In the following analysis, we will put our focus on the applications of business analytics that come with social science. Past studies of this dataset were focusing more toward the machine learning classification (e.g., to predict whether a guest will cancel their booking). But in this project, we will be focusing on time series to analyze macroscopic demand trends.

To achieve this goal, I applied two methodologies: First, the SARIMA(p, d, q)x(P, D, Q) model is applied based on the Box-Jenkins approach. Then, the machine learning approach, Neural Network Autoregression (NNAR), is applied to confirm/or provide a different interpretation of the result.

## 2. Data

The data for this project was downloaded from the Hotel booking demand dataset on Kaggle. It is originally from the article Hotel Booking Demand Datasets, written by Nuno Antonio, Ana Almeida, and Luis Nunes for Data in Brief, Volume 22, February 2019. It was subsequently cleaned and uploaded to Kaggle by Jesse Mostipak.

Data was obtained directly from the PMS databases' servers of two hotels (a city hotel and a resort hotel) in Portugal. It contains comprehensive booking information, including arrival dates, length of stay, and average daily rates.

Dataset Characteristics:

- Time range: From July 2015 to August 2017.
- Frequency: Data was originally recorded on a daily basis, but I aggregated it into a weekly frequency to prevent the sample size limitations that occur with monthly aggregation.
- Values: The primary value being analyzed is the total number of bookings made per week.
- Size of the data set: Contains 119,390 rows and 32 columns. Once it was aggregated into a weekly time series, the sample size becomes approximately 115 observations.
- Importance: Get a better understanding on how seasonal trends impact commercial revenue.
- Purpose: Practice data manipulation and apply predictive forecasting models to a practical business problem.

# 3. Methodology

## 3.1 SARIMA $(p, d, q) \times (P, D, Q)_s$ Model

The primary methodology applied is the Box-Jenkins approach to fit a Seasonal Autoregressive Integrated Moving Average (SARIMA) model. A SARIMA model captures both standard and seasonal dynamics in a time series and is denoted generally by the equation:

$$\Phi_P(B^s)\,\phi(B)\,\nabla_s^D\,\nabla^d x_t = \delta + \Theta_Q(B^s)\,\theta(B)\,w_t$$

where $w_t$ is the usual Gaussian white noise process.

The methodology follows three core steps:

- Plot the time series and apply differencing (ordinary $d$ and seasonal $D$) until the it achieves stationary. Estimate the AR $(p, P)$ and MA $(q, Q)$ terms by examining ACF and PACF plots.
- Estimate model parameters using Maximum Likelihood. Use `auto.arima()` to select the model that minimizes the corrected $AICc$.
- Use Ljung-Box test and ACF plots to analyze the residuals of the chosen model.

## 3.2 Machine Learning (Neural Network Autoregression)

As a secondary methodology, a Machine Learning approach was applied using Neural Network Autoregression (NNAR). Unlike SARIMA or traditional linear AR models that assume a linear relationship between past and future values, NNAR captures the trends and cycles in time-series data without requiring explicit feature engineering.

The NNAR model is trained using backpropagation to minimize the forecasting error. This method was chosen because it can be used to handle high-frequency seasonality, and it would be great to have it serve as a comparison to the traditional linear Box-Jenkins approach.

# 4. Results

```r
hotel_data <- read_csv("hotel_bookings.csv")
```

```
## Rows: 119390 Columns: 32
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (13): hotel, arrival_date_month, meal, country, market_segment, distrib...
## dbl  (18): is_canceled, lead_time, arrival_date_year, arrival_date_week_numb...
## date  (1): reservation_status_date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

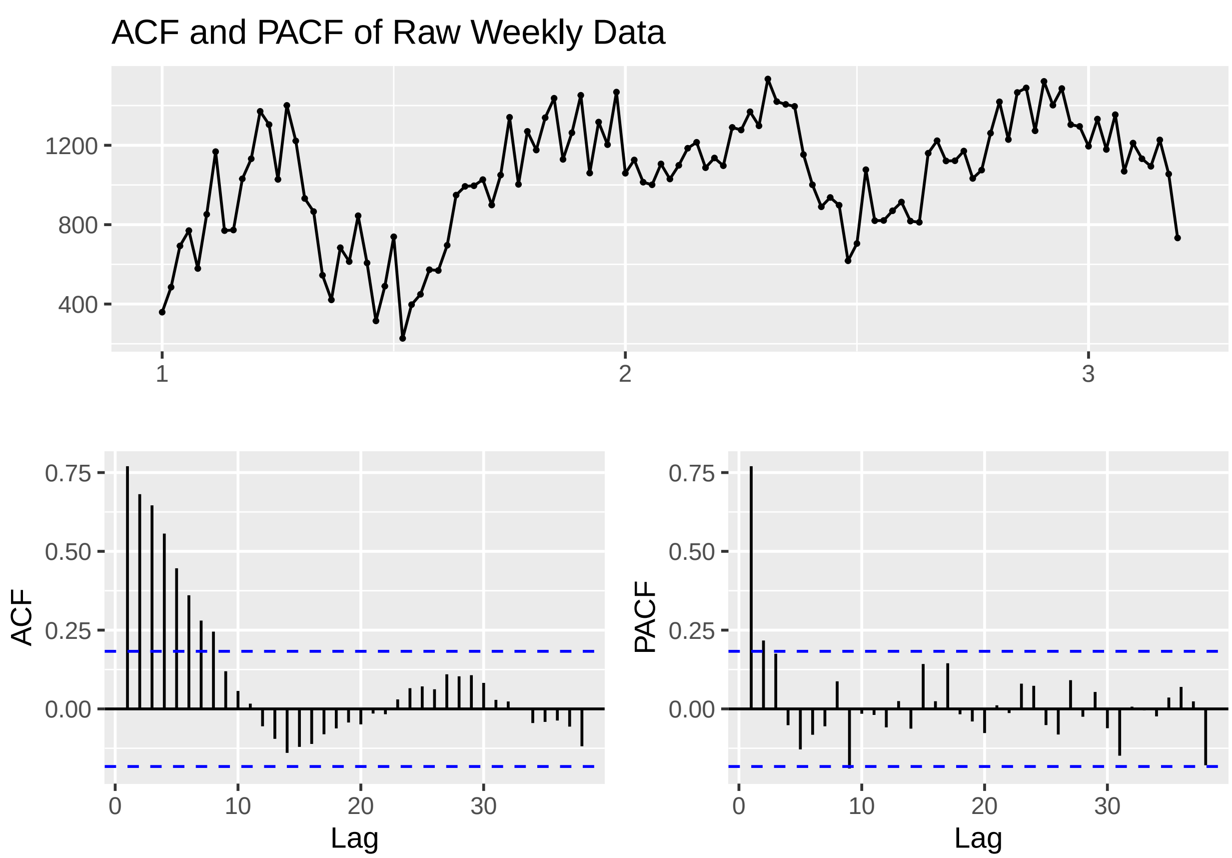
```r
weekly_bookings <- hotel_data %>%
  group_by(arrival_date_year, arrival_date_week_number) %>%
  summarise(total_bookings = n(), .groups = 'drop') %>%
  arrange(arrival_date_year, arrival_date_week_number)

ts_data <- ts(weekly_bookings$total_bookings, frequency = 52)
```

## 4.1 Results from SARIMA $(p, d, q) \times (P, D, Q)$

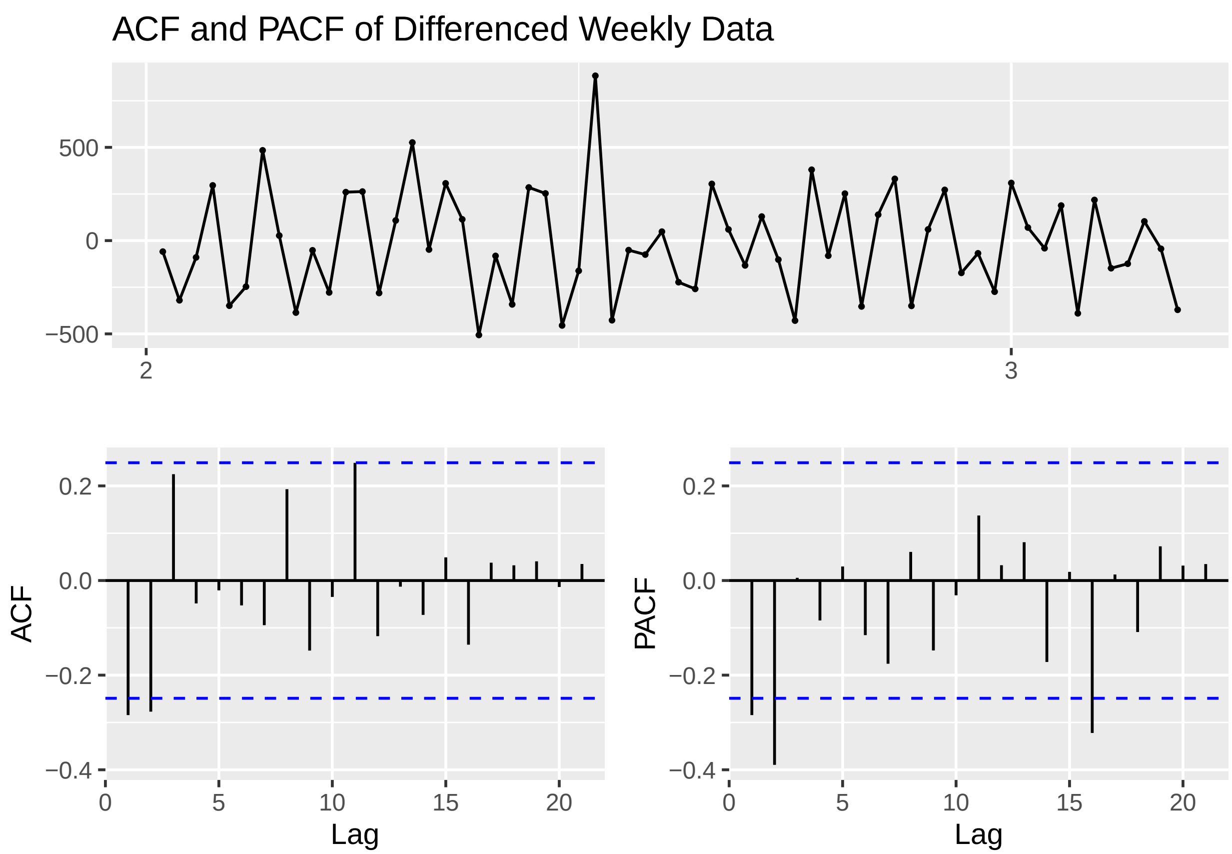
**Original and Transformed Data**

```r
ggtsdisplay(ts_data, main="ACF and PACF of Raw Weekly Data")
```



The initial plot of the raw weekly data reveals an upward trend during the first year, alongside significant oscillation corresponding to summer high seasons and winter low seasons. Because the ACF of the raw data trails off very slowly, it is evident that the data is non-stationary.

```r
diff_ts <- ts_data %>% diff(lag = 52) %>% diff()
ggtsdisplay(diff_ts, main="ACF and PACF of Differenced Weekly Data")
```




To achieve stationary, both a seasonal difference ($D = 1, s = 52$) and an ordinary difference ($d = 1$) were applied. The differences data oscillates around a constant zero mean.The ACF and PACF of the differenced data show negative spikes at early lags (specifically Lags 1 and 2). This indicates the presence of remaining autocorrelation.

### Model Selection and Diagnostics

The auto.arima() function was utilized to formally identify the optimal parameters by minimizing the $AICc$.

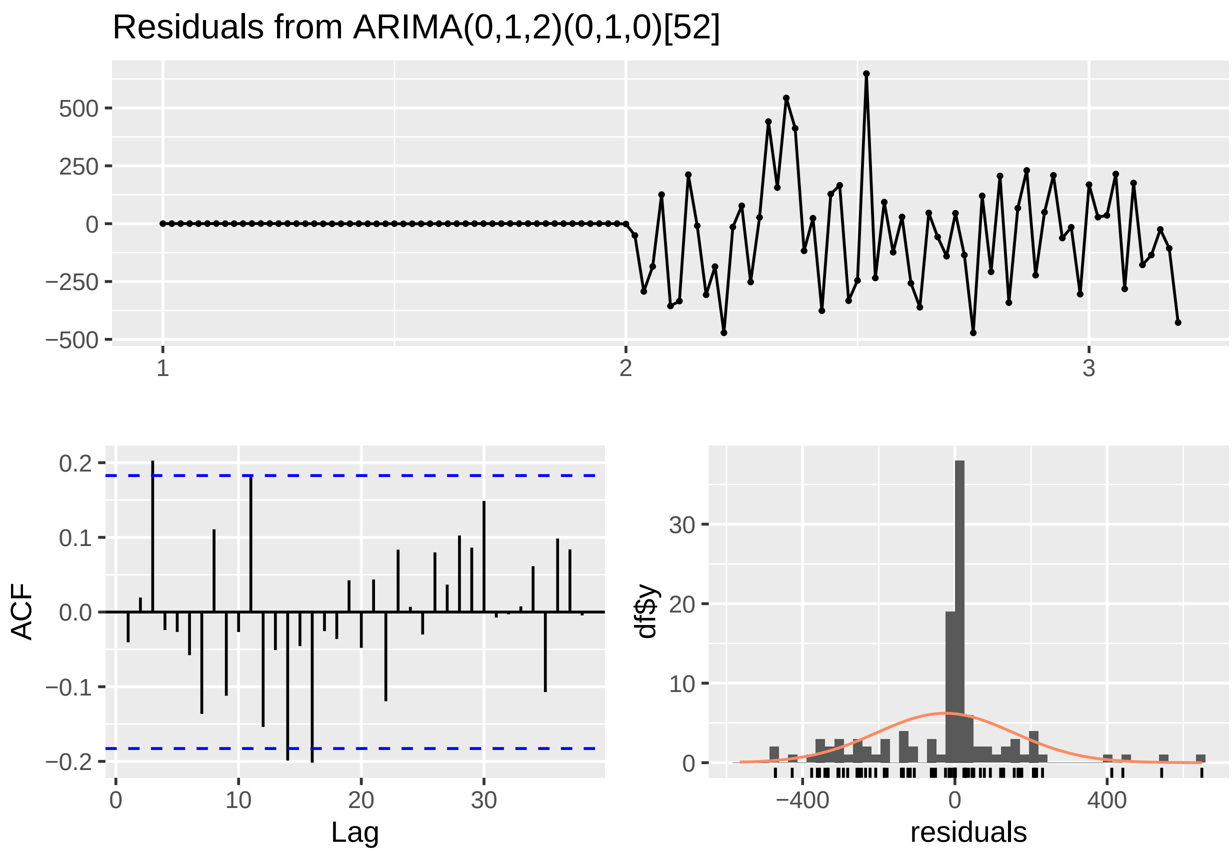
```r
optimal_model <- auto.arima(ts_data, stepwise=FALSE, approximation=FALSE)
summary(optimal_model)
```

```
## Series: ts_data
## ARIMA(0,1,2)(0,1,0)[52]
##
## Coefficients:
##           ma1      ma2
##       -0.4410  -0.3817
## s.e.   0.1374   0.1526
##
## sigma^2 = 62709:  log likelihood = -429.9
```

```
## AIC=865.81   AICc=866.22   BIC=872.19
##
## Training set error measures:
##                     ME     RMSE     MAE       MPE     MAPE      MASE
## Training set -25.56037 180.8807 107.119 -3.469174 10.02651 0.4160602
##                     ACF1
## Training set -0.04051507
```

The algorithm selected an ARIMA(0,1,2)(0,1,0)[52] model. This model utilizes 1 ordinary difference, 1 seasonal difference at lag 52, and 2 Moving Average terms ($\theta_1 = -0.4410, \theta_2 = -0.3817$).

```r
checkresiduals(optimal_model)
```

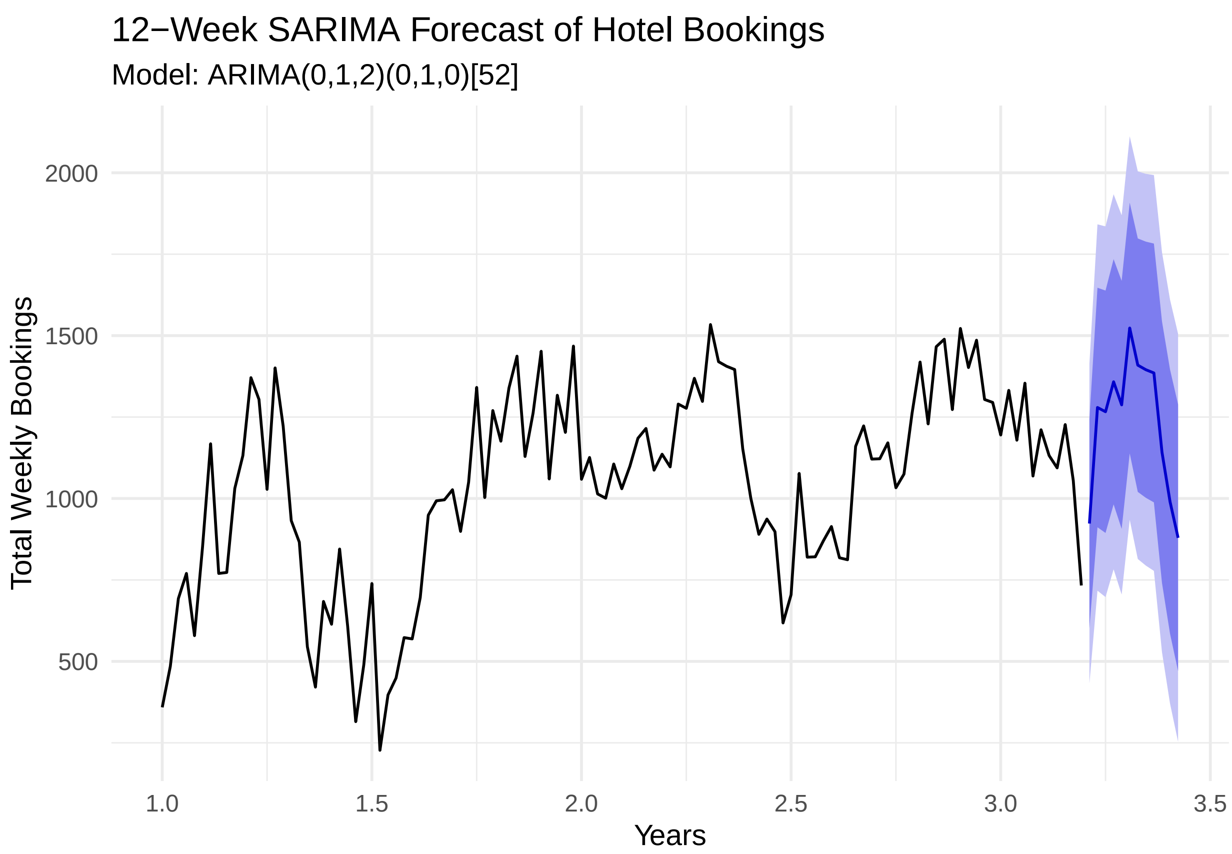



```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(0,1,2)(0,1,0)[52]
## Q* = 34.238, df = 21, p-value = 0.03417
##
## Model df: 2.   Total lags used: 23
```

The Ljung-Box test returned a p-value of 0.03417. We reject the null hypothesis that residuals are independently distributed. This is not very ideal, but it matches my expectation. Because weekly data is used alongside a seasonal difference of 52, the model consumes the entire first year's worth of data to compute the initial differences. Looking at the plot, we can see the first 52 weeks are flatlining at zero. It deflates the variance and inflates the test statistic. Given this phenomenon and the strong visual accuracy of the forecast, the model is still considered to be practically robust.

**SARIMA Forecast**

```r
final_forecast <- forecast(optimal_model, h=12)
autoplot(final_forecast) +
  labs(title="12-Week SARIMA Forecast of Hotel Bookings",
       subtitle = "Model: ARIMA(0,1,2)(0,1,0)[52]",
       x="Years",
       y="Total Weekly Bookings") +
  theme_minimal()
```




The plot details the 12-week forecast (blue line) with confidence intervals. The model successfully simulates the expected summer booking surge.

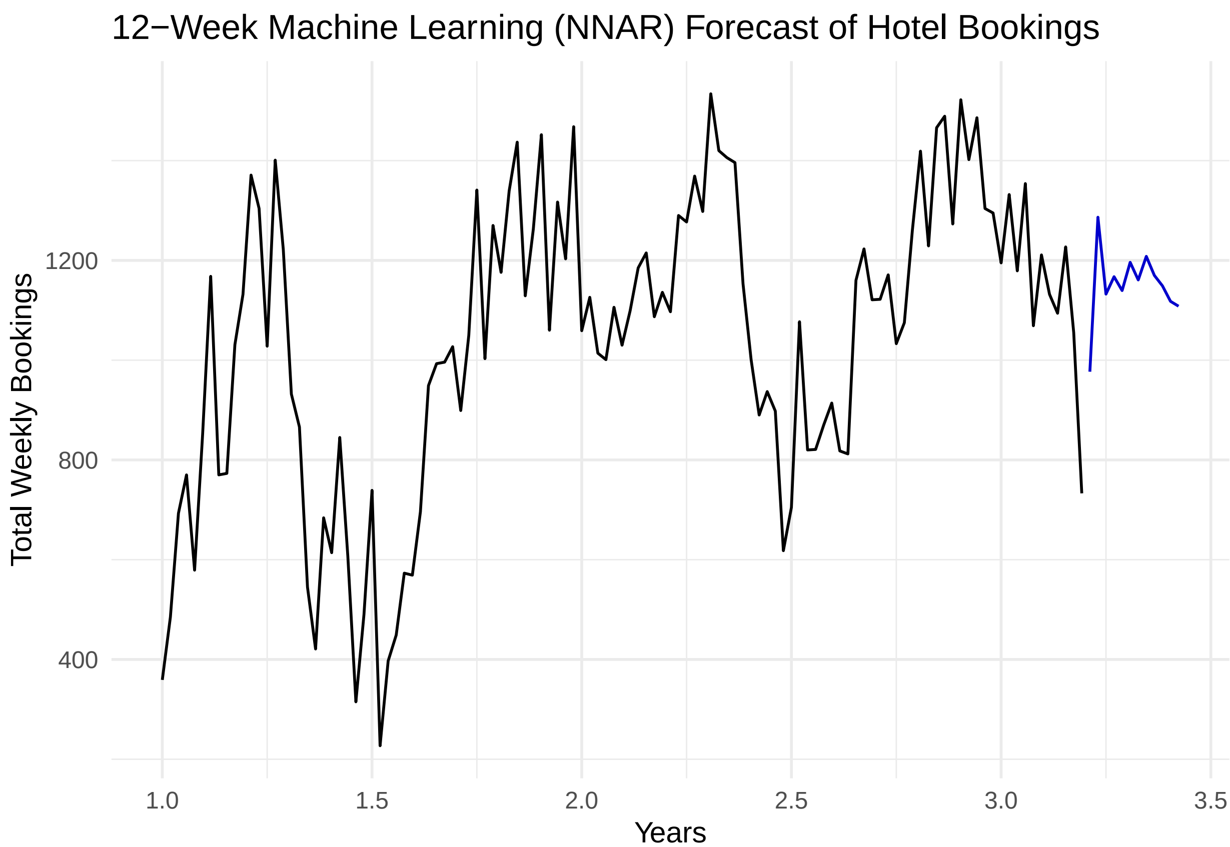
### 4.2 Results from NNAR

To validate the SARIMA findings and provide a non-linear comparative model, a Neural Network Autoregression model was fitted to the data. The `nnetar()` algorithm automatically determines the optimal number of lagged inputs and hidden nodes required to capture the 52-week seasonal patterns.

```
set.seed(921)
nn_model <- nnetar(ts_data)

nn_forecast <- forecast(nn_model, h=12)

autoplot(nn_forecast) +
  labs(title="12-Week Machine Learning (NNAR) Forecast of Hotel Bookings",
       x="Years",
       y="Total Weekly Bookings") +
  theme_minimal()
```




Interestingly, the Machine Learning forecast diverges from the findings of the SARIMA model. While SARIMA predicts an upward trend corresponding to the upcoming high season, the NNAR model's forecast remains relatively flat, hovering around 1100 to 1200 weekly bookings. The difference between these two models highlights a known limitation of neural networks in time series forecasting: NNAR requires a large sample size of data. Since there are only 115 weekly observations, the historical cycles are not enough for it to confidently assign heavy weights to the 52-week seasonal lag. This finding demonstrates that traditional parametric models like SARIMA are more reliable than unconstrained machine learning approaches for short-horizon and highly seasonal datasets.

# 5. Conclusion and Future Study

In this project, I successfully modeled and forecasted the weekly booking demand for two Portuguese hotels. Through exploratory analysis, a strong annual seasonal component was identified, reflecting the heavy cyclicality of the tourism industry.

Additionally, in the comparative analysis of SARIMA and NNAR, the chosen optimal model ARIMA(0,1,2)(0,1,0)[52] successfully parsed the non-stationary trend and seasonal patterns and predicted a summer booking surge that aligned with reality, while the non-linear NNAR model produced a relatively flat forecast that failed to anticipate the seasonal peak. This reveals that machine learning approaches like neural networks heavily rely on the sample size of the data even though they have higher flexibility. Overall, our finding demonstrates that for high-frequency seasonal data with short historical windows, traditional time-series models such as SARIMA offer greater advantages and exhibit more robust performance.

**Future Study:**

I think the future study could explore more about multivariate time series analysis. We can include more prediction variables, such as weather data, to improve forecast accuracy, and it might provide deeper insights into how the booking volume is impacted over time. In addition, finding and using a larger dataset that spans 5 to 10 years would be helpful for machine learning models to overtake traditional Box-Jenkins methods in capturing complex hospitality seasonality.

# References


- Antonio, N., Almeida, A., & Nunes, L. (2019). Hotel Booking Demand Datasets. Data in Brief, Volume 22.
- Mostipak, J. (2020). Hotel booking demand. Kaggle. Retrieved from https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand
- Shumway, R. H., & Stoffer, D. S. (2017). Time series analysis and its applications: With R examples (4th ed.). Springer.


# Appendix: R Source Code

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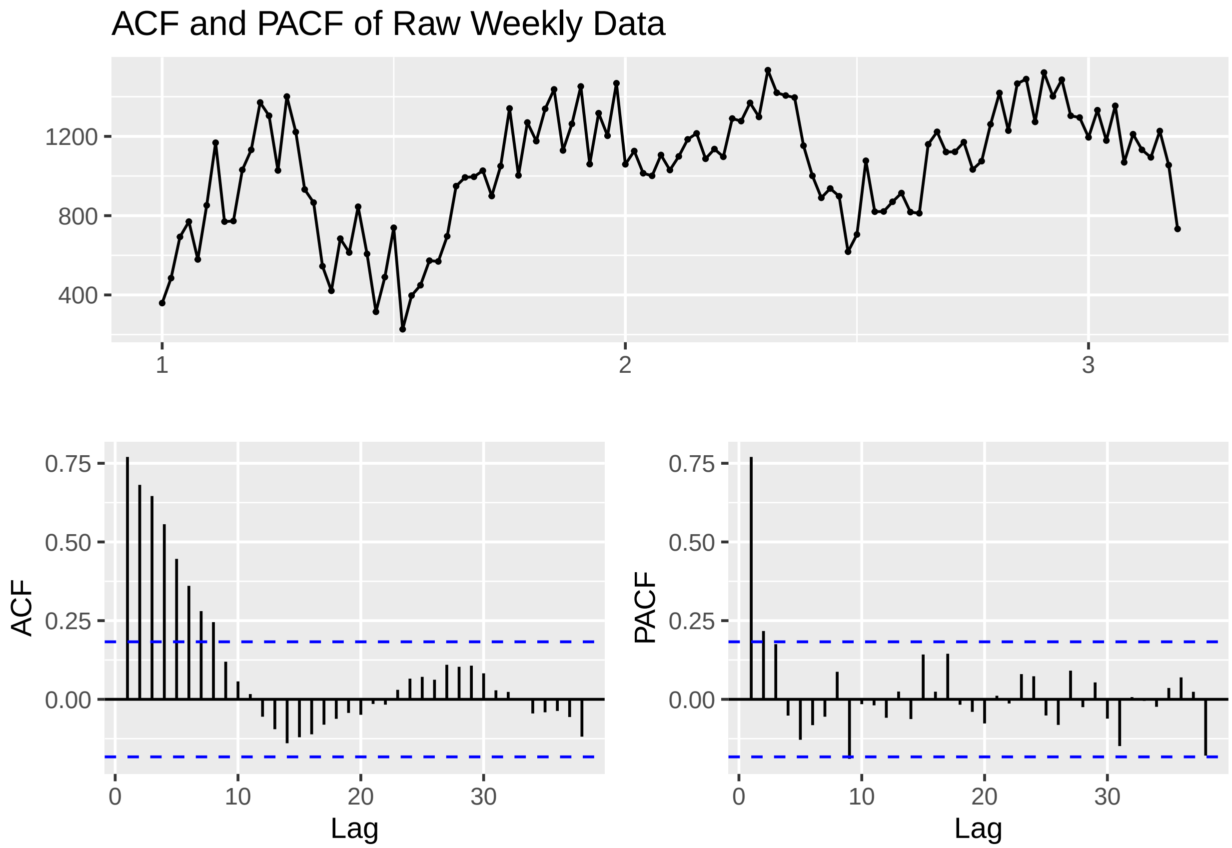


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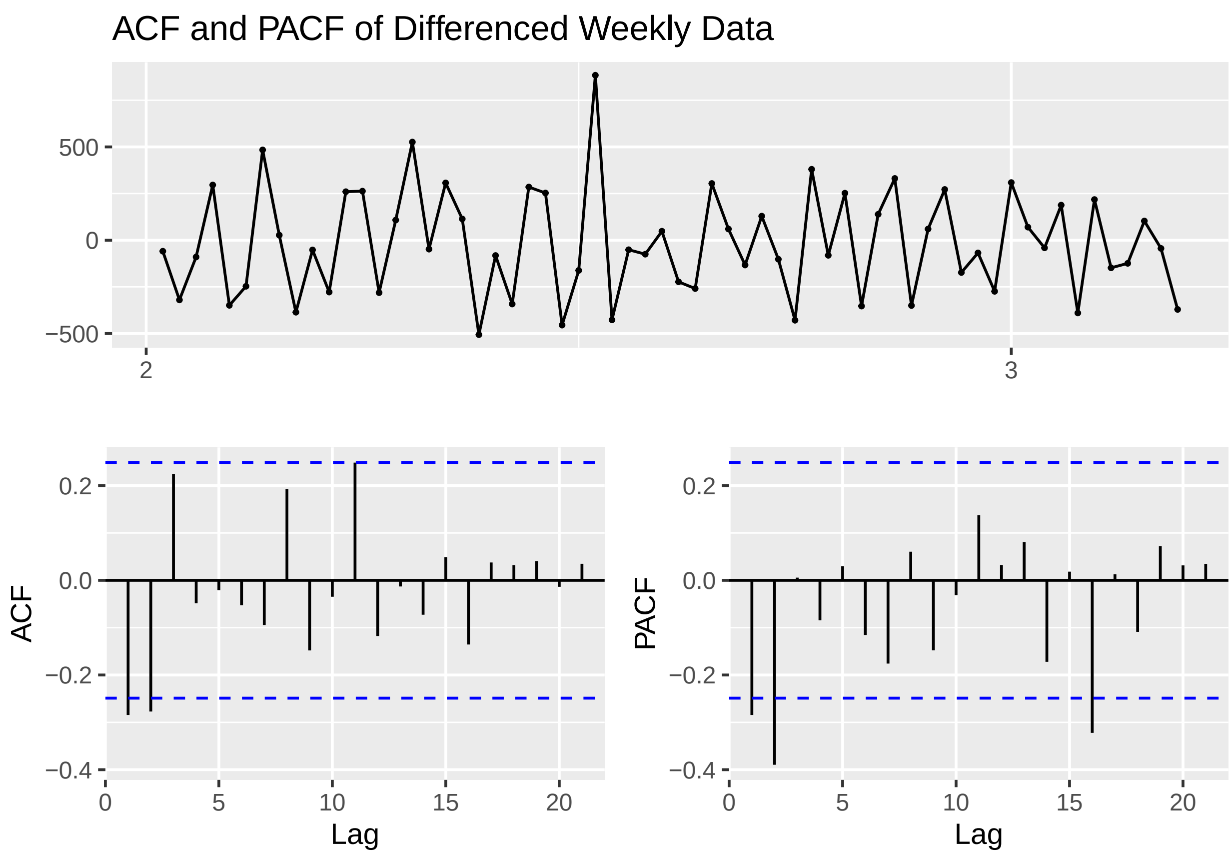


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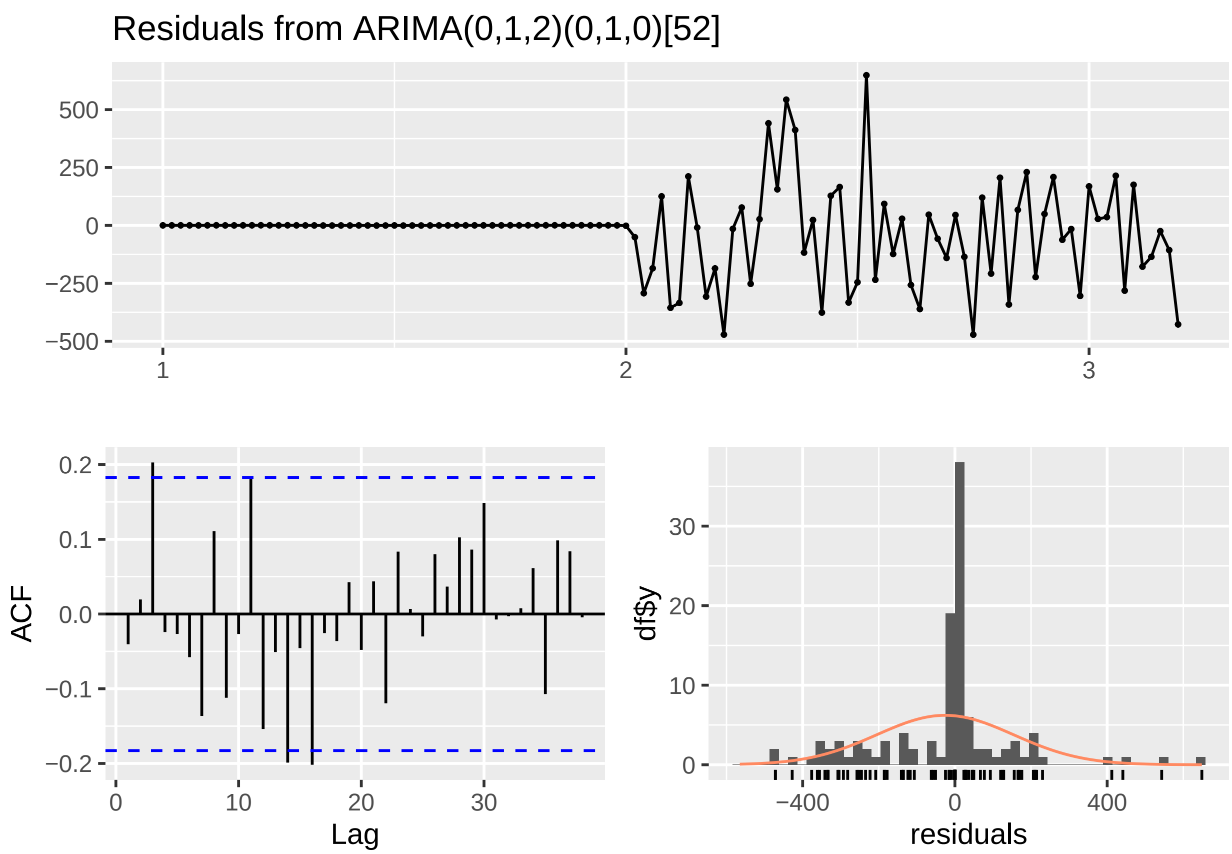


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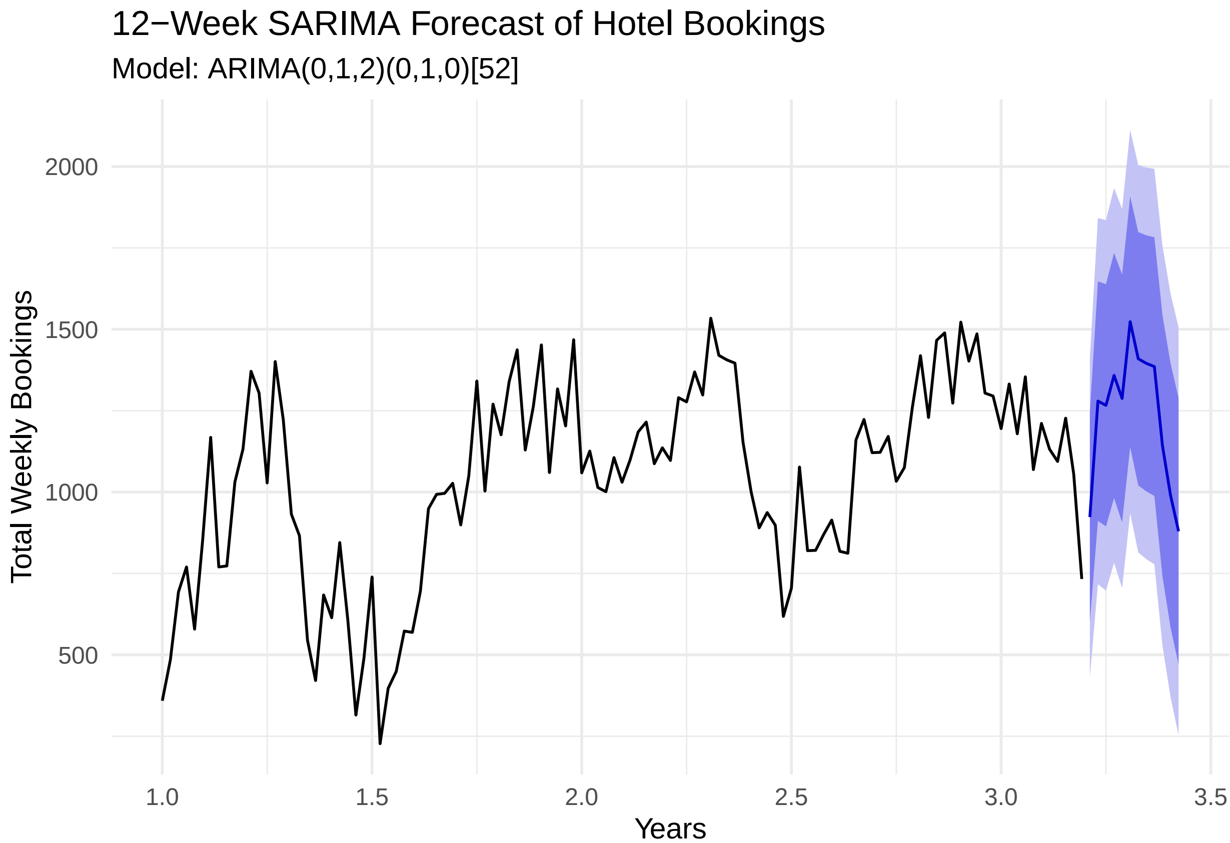


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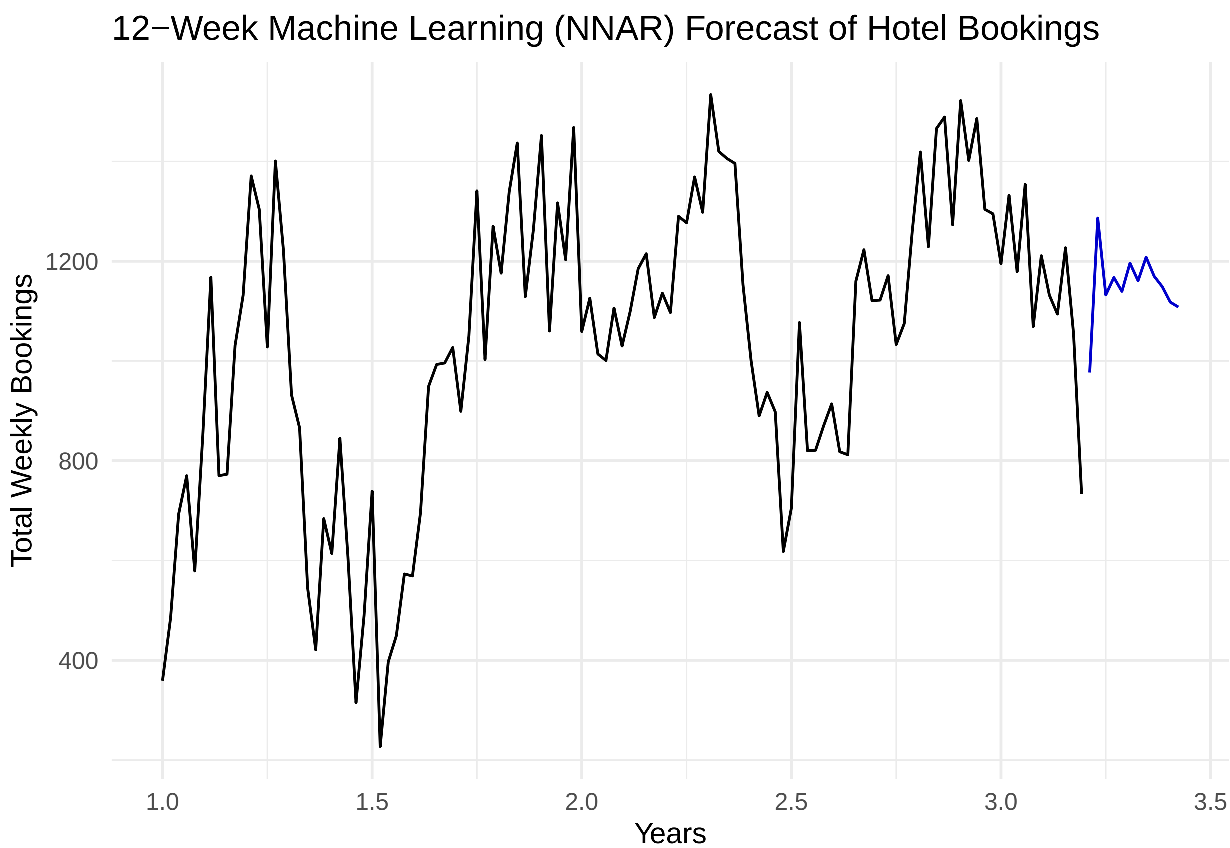
12−Week Machine Learning (NNAR) Forecast of Hotel Bookings
Total Weekly Bookings
1200
800
400
1.0
1.5
2.0
2.5
3.0
3.5
Years